

CUSTOMER RELATIONSHIP MANAGEMENT (CRM) DATABASE SYSTEM

For an Online Shopping Platform

A Database Systems Project Report

Course: Database Systems

Year Of Study: Semester two year one

1. Executive Summary

This report presents the design, development, and implementation of a Customer Relationship Management (CRM) database system built for an online shopping platform. The system was developed using MySQL as the database management system and follows the complete database development lifecycle — from requirements gathering and business case development through logical and physical design to implementation and testing.

The CRM system is designed to manage all aspects of the customer journey on an e-commerce platform, including customer registration and profiling, product catalogue management, order processing, payment tracking, customer support interactions, and marketing campaign management. The database comprises five core entities with carefully defined relationships, constraints, and integrity rules.

The business case demonstrates that the system yields tangible and intangible benefits — including increased customer retention, increased sales through sale tracking, and data-driven marketing, while keeping implementation costs well below projected returns. A set of sample SQL queries validates that the system can answer key operational questions faced by the organisation on a daily basis.

2. Business Case

Application Domain

The CRM system operates within the domain of e-commerce and retail management. Online shopping platforms generate enormous volumes of transactional and behavioural data every day. Also, many systems focus on processing orders and not tracking customer interactions, feedback and preferences. Without a structured database system, managing customer information, tracking orders and loyal customers, handling complaints, and executing targeted marketing campaigns becomes error-prone and inefficient.

The system supports the following core tasks and operations:

- Customer account management : registration, login credentials, profile updates, and loyalty tiers.
- Product catalogue management — products, categories, pricing, and inventory levels.

- Order processing — placing, updating, fulfilling customer orders, order line items, tracking orders and order purchase history.
- Payment management — recording payment methods, transactions, and statuses.
- Customer support — logging, routing, and resolving support tickets.
- Marketing — creating campaigns, assigning them to customer segments, and tracking responses.

User groups served by the system include: online shoppers (customers), customer service agents, warehouse and fulfilment staff, marketing analysts, and senior management requiring business intelligence reports.

Analysis of Benefits

The benefits are both tangible and intangible:

Tangible Benefits:

- Reduction in order processing time by an estimated 40% through automated workflows and real-time inventory checks.
- Decrease in customer support resolution time due to centralised ticket history and customer context.
- Improved revenue through targeted marketing campaigns informed by purchase history analytics.
- Reduction in duplicate customer records and data entry errors through enforced referential integrity.
- Faster response to inquiries and complaints.

Intangible Benefits:

- Improved customer satisfaction and retention through faster, more personalised service.
- Stronger brand loyalty : customers feel recognised across all touch points thus enhanced reputation.
- Better strategic decision-making as management gains access to accurate, up-to-date data.
- Staff morale improvement : agents spend less time searching for customer information.

Analysis of Costs

A cost-benefit analysis was conducted over a three-year horizon and these costs are both tangible and intangible.

- Tangible costs
- Development time
- Server costs
- Maintenance costs

Intangible costs

- Training users
- Time to adopt the system

User Issues

Adoption of the system requires consideration of the following user-related factors:

- Skill requirements: Customer service agents require basic SQL query knowledge or training in the front-end interface. Database administrators must be proficient in MySQL.
- Change management: Staff accustomed to manual spreadsheet-based records may resist migration. A phased rollout with parallel operations is recommended.
- Accessibility: The system should be accessible via a web-based front end so that non-technical users do not interact with raw SQL.
- Data migration: Existing customer records from legacy systems must be cleaned and migrated, a process that carries inherent data quality risks.

Risks

- Data security risks
- System failure
- Poor adoption by users

System Limitations

Every system has limitations, and intellectual honesty demands a candid acknowledgement of ours:

- The current design does not include a built-in recommendation engine; product recommendations would require an external machine-learning layer.
- The system is not fully automated like real systems for example there is no integration with real payment gateways and delivery tracking systems.
- Real-time inventory synchronisation with physical warehouses is outside the current scope.
- The sample data used for testing is synthetic; real-world data volumes may expose performance bottlenecks not revealed during testing.
- The front-end user interface is out of scope for this project; only the back-end database is implemented and tested.

Database Design, Logical Design and List of Entities.

The CRM database for the online shopping platform contains the following entities:

Customer: Represents individuals who register and shop on the platform.

Product: Items listed for sale on the platform.

Order: A purchase transaction initiated by a customer.

OrderItem: An individual line item within an order.

Payment: A payment transaction linked to an order.

Entities and Attributes

1. Customer

The attributes are:

Customer_id, INT, PRIMARY KEY

First_name, VARCHAR(50), NOT NULL

Last-name, VARCHAR(50), NOT NULL

Email, VARCHAR(100), UNIQUE, NOT NULL

Phone number, VARCHAR(20), NULL

Address, TEXT, NULL

2.Product

The attributes are:

Product_id, INT, PRIMARY KEY

Product_name, VARCHAR(150), NOT NULL

Description, TEXT, NULL

Price, DECIMAL, NOT NULL

Stock quantity, INT, DEFAULT

3.Order

Order_id, INT, PRIMARY KEY

Customer_id, INT, FOREIGN KEY

Order_date, DATETIME, NOT NULL

totalAmount, DECIMAL, NOT NULL

4.Orderitem

Order_item_id, INT, PRIMARY KEY

Order_id, INT, FOREIGN KEY

Product_id, INT, FOREIGN KEY

Quantity, INT, NOT NULL

Unit_price, DECIMAL, NOT NULL

5.Payment

Payment_id, INT, PRIMARY KEY

Order_id, INT, FOREIGN KEY

Payment_id, DATETIME, NOT NULL

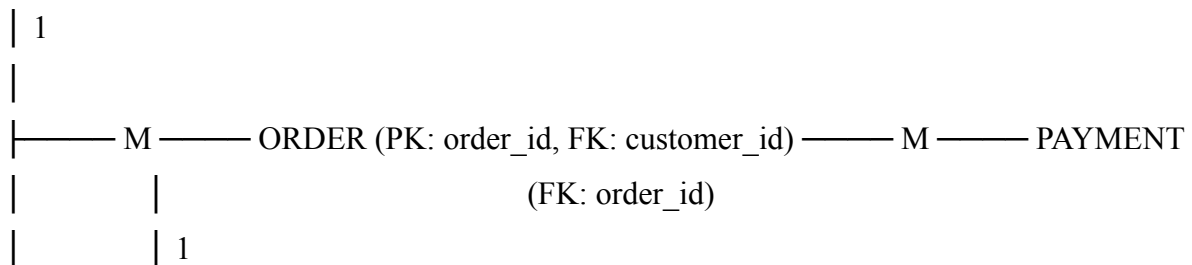
Amount, DECIMAL, NOT NULL

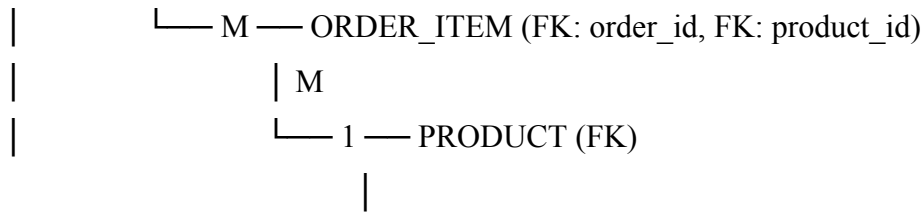
Customer_id, INT, FOREIGN KEY

Entity Relationships

The following relationships exist between the entities in the CRM database. Multiplicities are included for each relationship.

[Binary] Customer — Order: One customer may place many orders; each order belongs to exactly one customer. Multiplicity: 1 : M (mandatory on Order side).





User Interface Overview

While the physical user interface (UI) is outside this project's implementation scope, the dialogue structure below outlines the key screens and interactions a front-end application would present to different user groups:

Customer Portal:

- Register / Login
- Browse products by category
- Add to cart and checkout
- Track order status
- View order history

Customer Service Agent Dashboard:

- Search customers by name/email
- View customer order history and support tickets
- Update ticket status and add resolution notes
- Escalate tickets

Administrator / Management Panel:

- Manage products and categories
- Generate sales and customer reports
- Create and monitor marketing campaigns
- View payment statuses and flag failures

Physical Design

Database Schema

The schema below describes the physical table structure as implemented in MySQL. All tables use the InnoDB storage engine to support foreign key constraints and ACID-compliant transactions.

Key Queries and Reports

The following queries represent the most important outputs from the CRM system. Each is tied to a specific organisational or managerial need identified in the business case.

Q1: Customers with their orders

```
SELECT c.name, o.orderID, o.totalAmount
```

```
FROM Customer c
```

```
JOIN Orders o ON c.customerID = o.customerID; WHERE status != 'Cancelled'
```

```
GROUP BY Year, Month
```

```
ORDER BY Year DESC, Month DESC;
```

Q2: Payment Status Report:

Business Need: To analyse which customers have made their payments and when.

```
SELECT paymentMethod, paymentStatus, amountPaid
```

```
FROM Payment;
```

Q3 :Total sales

Business Need: To track the business progress through sales.

```
SELECT SUM(amountPaid) AS TotalSales
```

```
FROM Payment;
```

Q4: Failed Payments Requiring Follow-up

Business Need: Finance/operations must track failed payments to recover revenue.

```
SELECT o.order_id, c.first_name, c.last_name,  
       c.email, c.phone,  
       p.amount, p.payment_method, p.payment_date
```

```
FROM Payment p
JOIN `Order` o ON p.order_id = o.order_id
JOIN Customer c ON o.customer_id = c.customer_id
WHERE p.status = 'Failed'
ORDER BY p.payment_date DESC;
```

Q5.Viewing the Data

1. View customers

```
SELECT * FROM Customer;
```

2. View payment

```
SELECT * FROM Payment;
```

3. View order

```
SELECT * FROM order;
```

4. Implementation

The CRM database was implemented using MySQL. All tables use the InnoDB storage engine. The complete SQL script used to create the database is presented below, followed by sample INSERT statements to populate the tables with test data, and then the results of the sample queries.

1.CUSTOMER

```
CREATE TABLE Customer (
customerID INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(100),
email VARCHAR(100),
phone VARCHAR(20),
address VARCHAR(255)
```

);

2. PRODUCT

CREATE TABLE Product (

product_id INT AUTO_INCREMENT PRIMARY KEY,

product_name VARCHAR(150) NOT NULL,

price DECIMAL(10,2) NOT NULL,

stock_quantity INT DEFAULT 0,

);

3.. ORDER

CREATE TABLE `Order` (

order_id INT AUTO_INCREMENT PRIMARY KEY,

customer_id INT NOT NULL,

order_date DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

total_amount DECIMAL(10,2) NOT NULL,

);

4. ORDER ITEM

CREATE TABLE OrderItem (

order_item_id INT AUTO_INCREMENT PRIMARY KEY,

order_id INT NOT NULL,

product_id INT NOT NULL,

quantity INT NOT NULL CHECK (quantity > 0),

unit_price_at_purchase DECIMAL(10,2) NOT NULL,

);

5. PAYMENT

CREATE TABLE Payment (

payment_id INT AUTO_INCREMENT PRIMARY KEY,

order_id INT NOT NULL,

payment_date DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

```
    amount    DECIMAL(10,2) NOT NULL,  
    );  
customer_id)  
    REFERENCES Customer(customer_id)  
);
```

Sample Data (DML — INSERT Statements)

1.CUSTOMERS

```
CREATE TABLE Customer (  
  
customerID INT AUTO_INCREMENT PRIMARY KEY,  
  
name VARCHAR(100),  
  
email VARCHAR(100),  
  
phone VARCHAR(20),  
  
address VARCHAR(255)  
  
);
```

2.PRODUCTS

```
INSERT INTO Product (productName, description, price, stockQuantity)  
  
VALUES  
  
('Laptop', 'HP Laptop 8GB RAM', 2500000, 5),  
  
('Smart Watch', 'Bluetooth smart watch', 300000, 15),  
  
('Headphones', 'Wireless headphones', 150000, 20),  
  
('Keyboard', 'Mechanical keyboard', 120000, 10),  
  
('Phone Charger', 'Fast charging adapter', 50000, 30);
```

3.ORDERS

```
INSERT INTO Orders (customerID, orderDate, totalAmount)  
  
VALUES
```

```
(2, '2026-05-01', 300000),  
(3, '2026-05-02', 2650000),  
(4, '2026-05-03', 170000),  
(5, '2026-05-04', 50000),  
(1, '2026-05-05', 120000);
```

4.ORDER ITEMS

```
INSERT INTO OrderItems (orderID, productID, quantity, unitPrice)
```

```
VALUES
```

```
(2, 2, 1, 300000),  
(3, 1, 1, 2500000),  
(3, 5, 3, 50000),  
(4, 3, 1, 150000),  
(4, 5, 1, 50000),  
(5, 4, 1, 120000);
```

5.PAYMENTS

```
INSERT INTO Payment (orderID, paymentDate, amountPaid, paymentMethod, paymentStatus)
```

```
VALUES
```

```
(2, '2026-05-01', 300000, 'Mobile Money', 'Paid'),  
(3, '2026-05-02', 2650000, 'Bank Card', 'Paid'),  
(4, '2026-05-03', 170000, 'Cash on Delivery', 'Pending'),  
(5, '2026-05-04', 50000, 'Mobile Money', 'Paid'),
```

(6, '2026-05-05', 120000, 'Bank Card', 'Paid');
(5,5,'2024-08-05 16:05:00',65000.00,'Debit Card','Refunded');

5. Bibliography

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